Vehicle Yaw Stability Control Using Differential Braking

by

Philopatear Salib

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University of Waterloo

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Abstract

This paper presents a study on the use of differential braking for yaw stability control of a vehicle. A literature review is conducted to examine previous research in the field, which shows that yaw stability control is an important aspect of vehicle safety and handling. The literature review also highlights the use of differential braking as a method for yaw stability control and its advantages over other methods. Based on the literature review, a vehicle bicycle model is developed, which includes a torque input at the center of gravity. The bicycle model considers the vehicle's lateral dynamics and the effects of tire forces on the vehicle's yaw motion. The model considers the vehicle's speed and road conditions, which are important factors that affect yaw stability.

A yaw control system is also developed for the bicycle model, which uses differential braking as the control input. The system also includes a control algorithm that adjusts the braking torque based on the vehicle's speed and road conditions.

Simulations are conducted to evaluate the bicycle's response to various maneuvers with and without the proposed yaw control system. The results of the simulations show that the proposed yaw control system effectively improves the vehicle's yaw stability in various driving scenarios. In conclusion, this study presents a yaw stability control system using differential braking for a vehicle bicycle model. The proposed control system is able to improve the vehicle's yaw stability and handling in various driving scenarios. This research could be useful for the development of advanced driver-assistance systems (ADAS) and autonomous vehicles.

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I. Introduction

Various research studies have highlighted electronic stability control system importance in reducing crashes. Statistical analysis has shown that electronic stability control systems contributed in preventing about 40% of vehicle crashes due to loss of control [1]. Loss of directional control means that while a vehicle is on road, it is not able to follow the driver expected path. This usually takes place because of yawing or rotating of the vehicle due to fast maneuvering, hard steering, or road departure. Which can result in a subsequent rollover occurring on or off the road [2].

Different approaches, as differential braking and active steering systems, are implemented to provide vehicle's yaw control. Numerous studies have focused on improving the yaw control system performance by combining both systems. This integration was a trade-off in terms of the system complexity [3] [4] [5]. This study believes that differential braking control system only is able to provide a sufficient yaw moment to keep the vehicle under control.

Vehicle yaw stability control system is usually implemented as an active safety system to maintain the vehicle's handling and stability [6]. However, many research papers have addressed vehicle yaw stability control using differential braking as an essential approach for many other applications. It has shown remarkable results in lane-keeping assistance control [7] [8], collision avoidance [9], and redundancy for both electric power assisted steering in non-autonomous vehicles [6] and for steering systems in autonomous vehicles [10]. While vehicles having differential braking control have exhibited a great capability of achieving the desired yaw rate, it did not usually match the longitudinal response expected by the driver [11]. This has brought the attention to integrate it with active torque distribution [11] or with active aerodynamic control [12] with the aim of providing active control of both traction and yaw moment.

Diverse control algorithms are employed as an approach for the applications. The control strategy is usually based on an upper control and a lower controller. The upper controller calculates the desired yaw rate from a linear 2 DOF vehicle model, compares it to the actual yaw rate measured, and obtains the corrective yaw moment required to achieve the vehicle stability. While the lower controller act as a brake torque distributor according to the yaw rate angle and the amount of corrective yaw moment to be applied.

It is noticeable that most of the research papers which addressed the control algorithms of vehicle handling stability has utilized a linear bicycle model to calculate the desired yaw rate [7]- [9].The linear

model is verified that it is sufficient enough to convey accurate yaw rate and side-slip angle for control algorithms and research studies [13].

This report explores different control algorithms for upper and lower controllers in Section 2. While Section 3 represents the linear and non-linear models employed for vehicle and tire dynamics. Additionally, it includes the controller principle design and its architecture. Sections 4 and 5 represents, discusses, evaluates and interprets the simulation results, while pointing out possible future work for yaw vehicle control using differential braking.

II. Literature review

Since the 1990s, researchers have adopted the idea of controlling the vehicle yaw motion by controlling the distribution of the driving force generated at the left and the right wheels. This was referred to as direct yaw moment control or torque vectoring control [14]. The concept has been extended to spot the light on adjusting the distribution of braking forces to control the vehicle behavior through the generation of a yawing moment [15]. As early as those studies were carried out, their results have exhibited a dramatic improvement in the handling and the stability of vehicles.
Since then, this field has been extensively investigated and has constantly improved the capability of the vehicles to achieve the expected yaw rate by the driver. However, those vehicles did not usually match the desired longitudinal response. This has brought the attention to integrate differential braking control system with active torque distribution [11] or with active aerodynamic control [12] with the aim of providing active control of both traction and yaw moment.

Many developments have exploited differential braking capabilities for various applications. These developments managed to propose remarkable results in lane-keeping assistance control [7] [8], collision avoidance [9], and redundancy for both electric power assisted steering in non-autonomous vehicles [6] and for steering systems in autonomous vehicles [10].

The controller is designed to improve the performance of lane keep assist systems by using feedforward control to anticipate and correct for lane departure. The paper discusses the design and implementation of the controller, as well as the results of experiments that were conducted to evaluate its performance [4].

There are other references that links EPAS systems to the differential braking system such as [3]. The system is designed to improve the safety of EPAS systems by detecting and responding to failures in the EPAS system and using differential braking to maintain control of the vehicle. The paper discusses the design and implementation of the fail-safe control system, as well as the results of experiments that were conducted to evaluate its performance.

There are other papers that discusses the performance of several controllers for direct yaw moment, including PI, PD, and LQR controllers, and evaluate their performance in terms of stability,

robustness, and responsiveness. The paper discusses the results of simulations that were conducted to evaluate the performance of the controllers and provides a comparison of the controllers in terms of their performance and suitability for different applications [1].

Furthermore, similar models have been made that is quite similar to the work that will be done in the research such, "The 3-DoF bicycle model with the simplified piecewise linear tire model" (2013) presents a three-degree-of-freedom (3-DoF) bicycle model that incorporates a simplified piecewise linear tire model. The model is designed to simulate the dynamic behavior of a bicycle, including the effects of tire forces on the vehicle's motion. It describes the development of the 3-DoF model and the tire model, as well as the results of simulations that were conducted to evaluate the performance of the model. The paper also discusses the advantages of the simplified piecewise linear tire model over more complex tire models, such as its ease of implementation and computational efficiency [5].

Vehicle models utilized in differential braking studies are diversified. However, lateral, longitudinal and yaw motion are the most common interest between those studies, besides the wheel velocities [11] [16].

Although, tens of research papers have utilized a wide range of different control algorithms, most of which based their set point of the yaw rate on the linear bicycle model of the vehicle [7]- [9].
A computational simulation of a vehicle stability control (VSC) system based on a coordinated control algorithm that uses proportional-integral-derivative (PID) controllers and differential braking. The simulation is designed to evaluate the performance of the VSC system and its ability to improve vehicle stability. The paper discusses the results of the simulation and the advantages of using a coordinated control algorithm that combines PID controllers and differential braking [7].

Another example presents a self-tuning fuzzy proportional-integral-derivative (PID) controller applied to direct yaw moment control for vehicle stability. The controller is designed to improve the stability of a vehicle by controlling the yaw moment using steering, brakes, and powertrain. The paper discusses the design of the controller and the results of simulations that were conducted to evaluate its performance[8]. Another, presents a model-based design of an electronic stability control

(ESC) system for passenger cars. The paper discusses the use of CarSim and Matlab-Simulink to develop a model of an ESC system, and the use of the model to evaluate the performance of the ESC system. The paper also discusses the results of simulation and testing of the ESC system and the advantages of using a model-based design approach for the development of ESC systems [9].

All three papers are discussing control systems for vehicles, either for VSC, direct yaw moment control and electronic stability control (ESC). They all are discussing the design, implementation and results of the simulations conducted to evaluate the performance of the proposed systems. The first two papers are discussing the use of PID controllers and self-tuning fuzzy PID controllers in these systems, while the third one is discussing the use of model-based design approach to develop an ESC system.

The use of a direct yaw moment control system was also mentioned in several papers [14] and [12], which uses sensors such as steering angle sensors, yaw rate sensors, and lateral acceleration sensors to measure the vehicle's motion, and applies the appropriate torque to the wheels to correct any unwanted yaw motion. The paper conducts simulations and vehicle testing to evaluate the performance of the direct yaw moment control system and its ability to improve vehicle stability and maneuverability.

The results showed that the direct yaw moment control system improved vehicle maneuverability by reducing the yaw angle deviation and yaw rate deviation during a lane change maneuver and also showed improved vehicle stability during a high-speed cornering and a step-steer maneuver.

Overall, the paper concludes that the use of direct yaw moment control improves vehicle maneuverability by reducing unwanted yaw motion, improving vehicle stability, and increasing the precision of the vehicle's motion.

This shows that the recent searches conclude that a linear model is verified that it is sufficient to convey accurate yaw rate and side-slip angle for control algorithms and research studies [13].

The papers also investigate various braking strategies and compare their effectiveness in maintaining vehicle stability during different driving scenarios. They also analyze the impact of different braking strategies on tire wear and vehicle dynamics. The results of the study indicate that the use of

differential braking can significantly improve vehicle stability and may be a useful addition to traditional stability control systems [12].

The differential braking system has then been adjusted and the parameters were altered over the decades in order to reach the highest efficiency of the braking system.

The papers show various variations in the input parameters as well as the PID control logic, there are a lot of parameters when working with a 4-wheel vehicle that’s why it’s quite simpler to use a bicycle model which our research will focus on as well as use PID control logic which has been proved to give the best outcomes and results as shown in paper [10]-[14].

Many vehicle handling stability control algorithms use a linear bicycle model to calculate the desired yaw rate. This model represents the vehicle as a simplified "bicycle" with two wheels and uses linear equations to describe the dynamics of the vehicle. The desired yaw rate is calculated based on inputs such as steering angle and vehicle speed is used to control the vehicle's stability and handling characteristics. The model's simplicity allows for quick and efficient calculations, making it well-suited for real-time control systems.

III. Modeling and Methodology

A. Introduction to the model

The bicycle model is modeled as 3 sub-systems, one for the Equation of Motion (EOM) another for the slip angle calculator and a final one for the tire forces calculator. The EOM sub-system describes the motion of the bicycle, the slip angle calculator sub-system calculates the slip angle of the wheels, and the tire forces calculator sub-system calculates the tire forces acting on the wheels. These sub-systems work together to model the dynamic behavior of the bicycle.

The EOM is typically formulated using the Newton-Euler equations of motion, which express the relationship between the forces acting on the bicycle and its acceleration, models the vehicle equation of motions with inputs of rear lateral force and front lateral force and outputs the angular velocity (omega), angular displacement (theta) and lateral velocity (Vy). The omega and Vy are then used as inputs for the slip angle calculator to find the rear slip angle and front slip angle and finally both slip angles are used as inputs for the tire forces calculator to find the rear lateral force and front lateral force to loop the model again. The obtained slip angles are then used to find the lateral force.

These three subsystems collaborate to simulate the dynamic behavior of the bicycle. The EOM subsystem describes the motion of the bicycle, while the slip angle calculator calculates the slip angle. The slip angle of the wheels sub-system offers information, and the tire forces calculator sub-system gives information on the tire forces acting on the wheels.

B. Bicycle Model:

As mentioned above the bicycle model was modeled as 3 subsystems: EOM, Slip Angle Calculator, and tire forces calculator.

C. Equation of Motion:

Modeling of the vehicle in a bicycle model requires two equations of motion. The first equation represents the movement of the vehicle in the lateral direction and the second represents the vehicle’s

yaw motion. Another equation was added to represent the braking force on the vehicle which will later help in controlling the vehicle's yaw moment. The equation representing the braking force was implemented by first determining the maximum braking force that is obtained by multiplying the vehicle mass and the gravitational acceleration coefficient by the coefficient of friction. By dividing the output by 2, the maximum braking force per wheel is obtained. The braking force will be either equal to the maximum or minimum braking force depending on the braking input. The braking force will then be multiplied by the half of the track width to obtain the applied torque that will be added to the following equations.

The 2 equations are represented as follows:

$\Sigma Fy = m * a$

Where: Fy: Lateral Force m: Vehicle mass a: Vehicle Acceleration

$\Sigma Mz = I * \alpha$

Where: Mz: Yaw Moment I: Vehicle Inertia α: Vehicle Angular Acceleration

The two equations will be broken to the following forms: $Yyf + Yyr = m * a$

$Yyf * l1 - Yyr * l2 = I * \alpha$

To model the two equations above, both lateral forces were left as values to be determined from the rest of the model. Adding the forces and dividing them by the mass yields the value of the acceleration. By integrating the acceleration, the velocity is obtained. In the last equation the L1 & L2 were obtained from researching the internet for different and common vehicle parameters; the vehicle inertia was estimated by multiplying the weight of the vehicle by L1 & L2. The angular acceleration of the vehicle was obtained by dividing the obtained result of subtraction by the vehicle inertia.

The omega and theta were obtained by integration. From the EOM, the inputs were the lateral forces and the outputs were the lateral velocity, omega and theta.

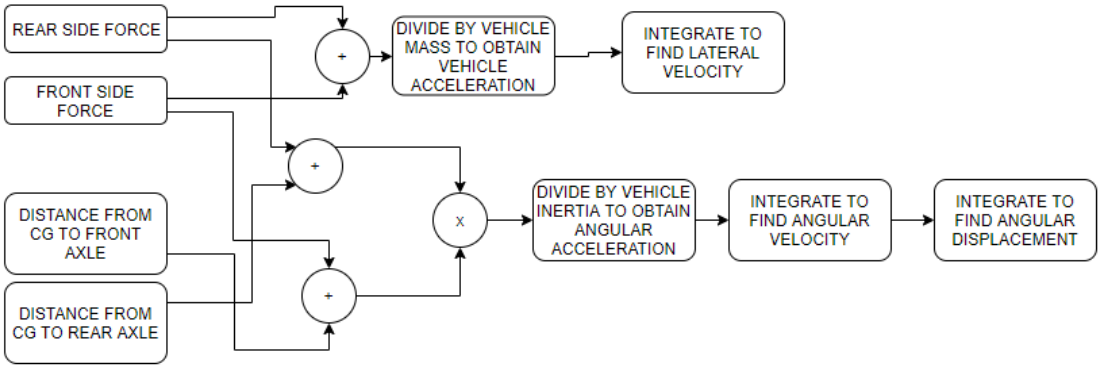


Figure 1 Demonstrates an overview of the bicycle system

D. Slip Angle Calculator:

The purpose of the slip angle calculator is to estimate the front and rear slip angles of the car. The slip angle calculator can be implemented in the Simulink model using a MATLAB function block. The function block takes the inputs of the velocity and acceleration of the bicycle and the tire forces, and uses them to calculate the slip angle.

The slip angle calculator is an important component of the bicycle model because it allows for a better understanding of the stability of the vehicle. It helps to determine the stability of the vehicle during different maneuvers and at different speeds and can be used to develop a yaw control system for the bicycle model.

According to the kinematics of the bicycle model, the front wheel is assumed to have a side force, tractive force and a resultant force with a slip angle between it and the longitudinal force. Also, a steering input is added as the front wheel is assumed to be steerable. The rear wheel has a front and side forces and resultant force with a slip angle between it and the longitudinal force. While the vehicle body is assumed to have a longitudinal force, side force and a side slip angle for the vehicle. All are assumed to act at the CG of the vehicle. To find the slip angle of the front wheel via trigonometric solution, the following equation was obtained:

$tan{-}1\ (Vy + wlf\ Vx) - \delta$

Vy: Lateral Velocity, w: Angular Velocity, lf: Wheel Base, Vx: Longitudinal Velocity, δ: Steering Input

To find the slip angle of the rear wheel via trigonometric solution, the following equation was obtained:

$tan{-}1\ (Vy - wlr\ Vx)$

Vy: Lateral Velocity, w: Angular Velocity, lr: Wheelbase, Vx: Longitudinal Velocity

It should be noted that the bicycle model always assumes that the longitudinal velocity is always constant. However, this model is associated with another longitudinal dynamics model which can interpret the longitudinal velocity accurately thus the model is modeled transiently instead of a steady state model. The steering input is the beginning of the model as the model results are affected by the steering input and the results change accordingly. From the slip angle calculator, the inputs were the lateral velocity, omega, distance from the CG height to the front of the vehicle and the distance from the CG height to the rear of the vehicle.

E. Tire Forces Calculator:

Tire force calculator is a tool that is used to calculate the forces acting on a vehicle's tires, which includes the vertical load, longitudinal force, and lateral force. It is used to understand the behavior of a vehicle's tires under different driving conditions, which can be used to improve vehicle performance, handling, and safety.

The vertical load is the force that is exerted on the tire perpendicular to the ground and is determined by the weight of the vehicle and its passengers. The longitudinal force is the force that is exerted on the tire in the direction of the vehicle's motion and is determined by the vehicle's acceleration and braking.

The lateral force is the force that is exerted on the tire in the direction perpendicular to the vehicle's motion and is determined by the vehicle's cornering and turning.

To find the lateral force, the front and rear slip angles are multiplied each by the cornering stiffness to obtain an estimated lateral force which is later used as feedback to the bicycle model as an input to the equation of motion.

F. Yaw Rate Control:

Yaw rate control is a technique used to improve the stability and handling of a vehicle by controlling the rotation rate of the vehicle about its vertical axis, also known as the yaw rate. The yaw rate is the rate at which the vehicle is turning, and it is measured in degrees per second.
The yaw rate has been measured; the control system uses this information to generate control inputs that will bring the yaw rate back to the desired range. This is typically done using a feedback control algorithm, In this case, proportional-integral-derivative (PID) controller. The PID controller compares the measured yaw rate to the desired yaw rate and generates control inputs that will bring the yaw rate back to the desired range.

To find a setpoint for the brakes to act on, two changes were made. While the vehicle was modeled as a bicycle with a 2 DOF model, another subsystem was designed with an input as the braking force applied to an assumed vehicle track that will be added to the equations of motion. The braking force is assumed to be equal to zero under normal conditions or if the ideal radius and the actual radius of the vehicle is equal. Thus, it can be assumed that the vehicle will neither oversteer nor understeer. However, if the vehicle starts to have a difference between the ideal radius and the actual radius then the PID controller shall try to actuate by applying a braking force that can equalize both radii. After calculating the ideal and actual radii, the difference between them, if not equal, will be the input to the PID controller which will then work by applying a braking force that will be added to the equation of motion in efforts to bring both radii to an equal value.

Determination of the Ideal Radius is through dividing the wheelbase by the input steering parameter. For the actual radius, the wheelbase will be divided by the addition of the input steering parameter, rear slip angle and the front slip angle.

Yaw rate control systems, along with other stability control systems like as lateral stability control and traction control, can be used to improve a vehicle's overall stability and handling. A yaw rate control system, for example, can be used in conjunction with a lateral stability control system to increase a vehicle's stability during high-speed maneuvers.

G. Transformation Matrix:

To study the forces on the ground rather than the vehicle a transformation matrix was done using a MATLAB function inside of the Simulink Model. Same as the following image

$$\mathbf{R}_Z(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 & 0 \\ \sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The transformation matrix can be implemented in the Simulink model using a MATLAB function block. The function block can take the inputs of the forces acting on the vehicle, such as the longitudinal force and the lateral force, and use them to calculate the forces on the ground.
The use of this transformation matrix can give a better understanding of the forces acting on the ground and how they affect the vehicle's stability and performance. The graphical readings generated by this model can show the forces on the ground in real time and how they change over time during different maneuvers, with or without a controller at different speeds and road conditions.
The script was written using MATLAB as follows:

```
function arrayOfOutputs = fcn(arrayOfInputs, angleOfRotation)
tf = [cos(angleOfRotation) -sin(angleOfRotation);sin(angleOfRotation) cos(angleOfRotation)];
arrayOfOutputs = inv(tf)*arrayOfInputs;
```

H. Vehicle Controller Design:

As previously explained, in order to maintain and control the vehicle yaw rate, a braking input was added to the model which will give an output of zero should the vehicle ideal and actual radii be equal. The previous condition will not cause the vehicle to oversteer or understeer, hence the need for a controller should not be valid. On the other hand, if the actual and ideal radii of the vehicle are not equal, which is the actual case that will happen almost all the time, a controller intervention should

happen. The controller shall compare the ideal and actual radii and determine the difference between them as an error value that needs to be zero. In order to determine the vehicle actual radius the following equation was used:

$$Actual\ Radius\ = \frac{Wheel\ Base}{Wheel\ Angle\ +\ Rear\ Slip\ Angle\ +\ Front\ Slip\ Angle}$$

The above equation was modeled through simulink in addition to the ideal radius equation which states:

$$Ideal\ Radius\ = \frac{Wheel\ Base}{Wheel\ Angle}$$

These two outputs were compared to determine if the PID controller should be active or not. If both are equal then the output will be zero in order to deactivate the PID controller and send the minimum braking force (0) to the wheel. On the other hand, if the difference between both signals are not equal, the difference between them will be the error value input to the PID controller. The PID controller had been tuned using the auto-tune feature of the simulink block while focusing on a robust and a fast response in order to make the error value equal to 0. The output will then be an input to the braking force which acts as an actuator to the model to control the vehicle yaw rate.

The PID auto tuner resulted in these following parameter as the tuning parameters to the controller:

Kp: -765.564265691975
Ki: -781534.171021295
Kd: -0.0269112999506061

Filter Coefficient (N): 5836.60149931409

IV. Simulation Results and Discussions

The result section is divided into two parts the graphical results demonstrated with out PID control and the results when the PID control parameters was added to it.When using a PID (Proportional-Integral-Derivative) controller on a bicycle model, the graphical readings will show a more stable and controlled response compared to not using a PID controller.

A PID controller is a feedback control system that uses the error between the desired output and the actual output to adjust the control inputs. In the context of a bicycle model, the PID controller can be used to control the yaw rate, which is the rate at which the bicycle rotates around its vertical axis.

When using a PID controller, the graphical readings will show a decrease in the yaw rate error over time. This means that the yaw rate of the bicycle will converge towards the desired yaw rate, resulting in a more stable and controlled response. The graphical readings will also show a smooth transition between the desired yaw rate and the actual yaw rate, with minimal overshoots and oscillations.

On the other hand, if a PID controller is not used, the graphical readings will show a larger yaw rate error and a less stable and controlled response. The yaw rate of the bicycle may not converge towards the desired yaw rate, and the graphical readings will show larger overshoots and oscillations

A. Model Response Without PID

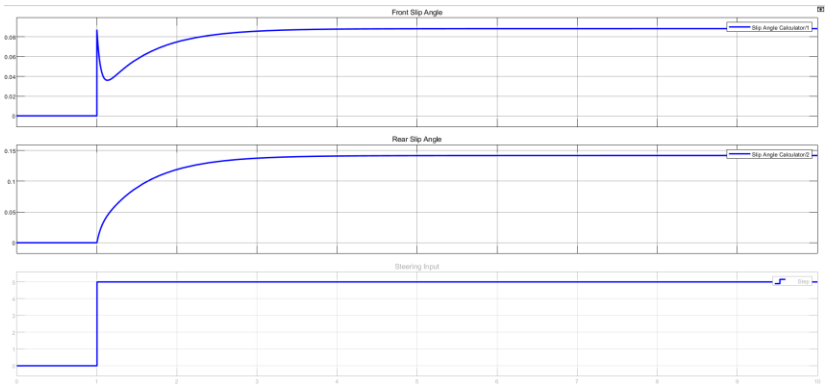


Figure 2:Steering Input of 5 degrees, Outputs: Front & Rear Slip Angles

This is the normal response of a rear and front slip angle when obtained through a bicycle model. This result has been cross checked with a textbook named, Theory of Ground Vehicles

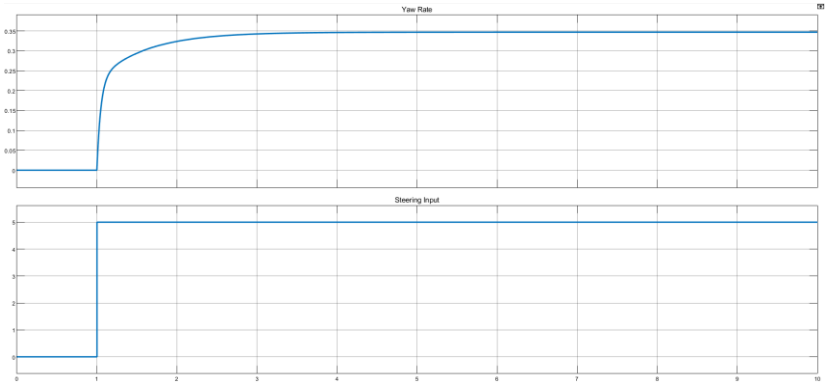


Figure 3: Yaw Rate vs. Steering Input at 5 degrees

This is the normal response of a yaw rate when obtained through a bicycle model. This result has been cross checked with a textbook named, Theory of Ground Vehicles.

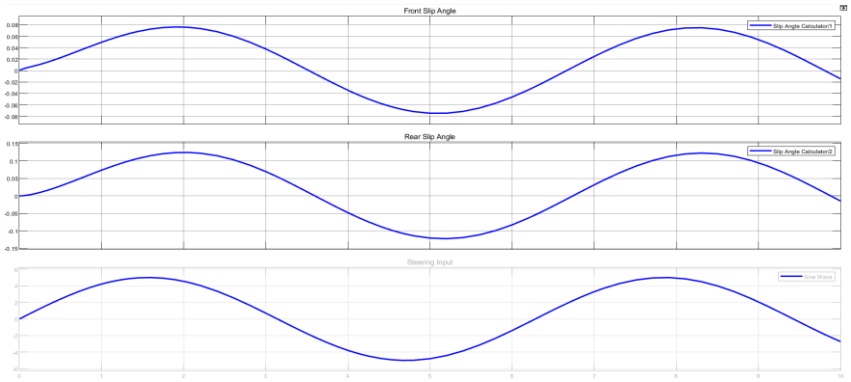


Figure 4 Slalom steering input, Outputs: Front and Rear Slip Angles

This is the normal response of a rear and front slip angle when obtained through a bicycle model and the input is represented as a slalom. In a bicycle model of a vehicle, the normal response of the rear and front slip angles when the input is represented as a slalom maneuver would depend on the specific parameters of the model and the conditions of the road. Typically, the slip angle is defined as the angle between the direction of motion of the wheel and the direction of the tire's actual path. In a slalom maneuver, the driver is turning the steering wheel rapidly from one side to another, causing

the vehicle to move laterally. This result has been cross checked with a textbook named, Theory of Ground Vehicles.

When a vehicle is in a steady-state turn, the slip angle at the front wheel is typically greater than the slip angle at the rear wheel. This is due to the fact that the front wheel is responsible for steering the vehicle, and as such must be able to rotate at a greater angle in order to change the direction of the vehicle.

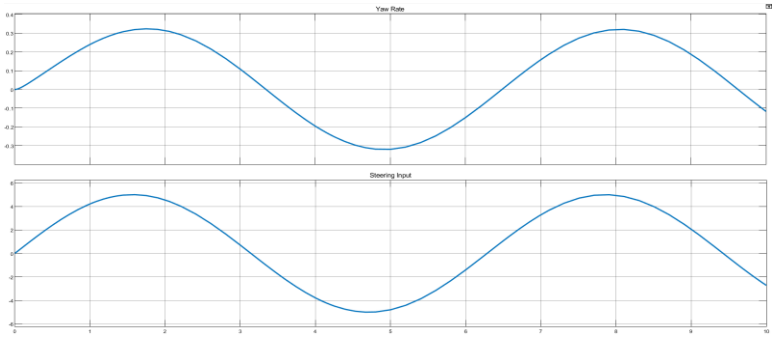


Figure 5 Yaw Rate vs Steering input as Slalom

This is the normal response of a yaw rate with a slalom input when obtained through a bicycle model. This result has been cross checked with a textbook named, Theory of Ground Vehicles.

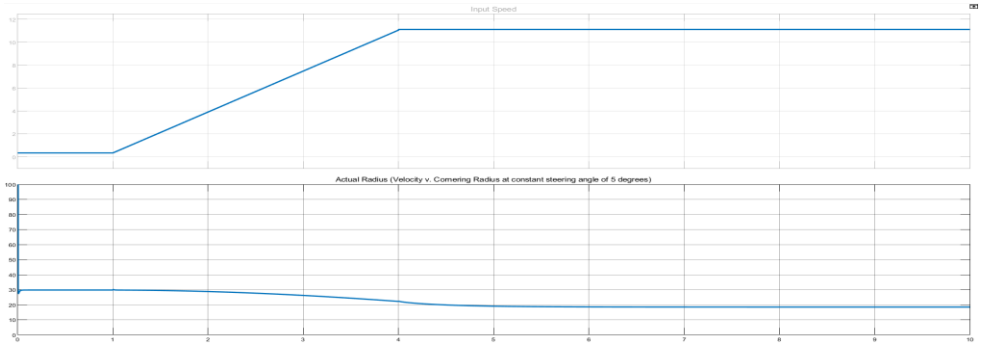


Figure 6 Transient Response of speed vs Actual Radius with constant steering of 5 degrees

For this result, the bicycle model is modeled in a steady-state environment with the speed assumed to be constant throughout the model. The actual radius behavior complies with the equation even though the model is simulated in a transient environment. In this case, the bicycle model is modeled in a steady-state environment, meaning that the speed of the bicycle is assumed to be constant throughout the model. This simplifies the model and allows for a better understanding of the forces acting on the

bicycle. However, in reality, the speed of the bicycle is not always constant and can change during different maneuvers. Therefore, the model is simulated in a transient environment, where the speed can vary over time.

Despite this, the actual radius behavior of the bicycle still complies with the equation, even though the model is simulated in a transient environment. This implies that the forces operating on the bicycle, such as the longitudinal and lateral forces, are still compatible with the equation and can be used to predict the behavior of the bicycle.

The use of the steady-state assumption allows for a better understanding of the forces acting on the bicycle in a controlled environment, but it is important to note that the results may not be directly applicable to real-world scenarios where the speed is not constant.

B. Model Response with PID:

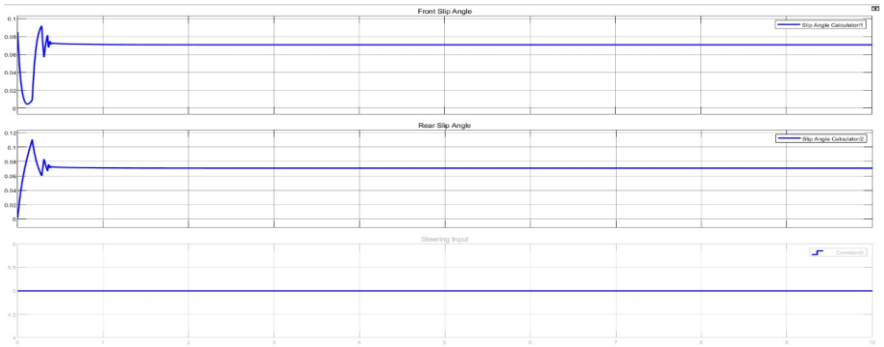
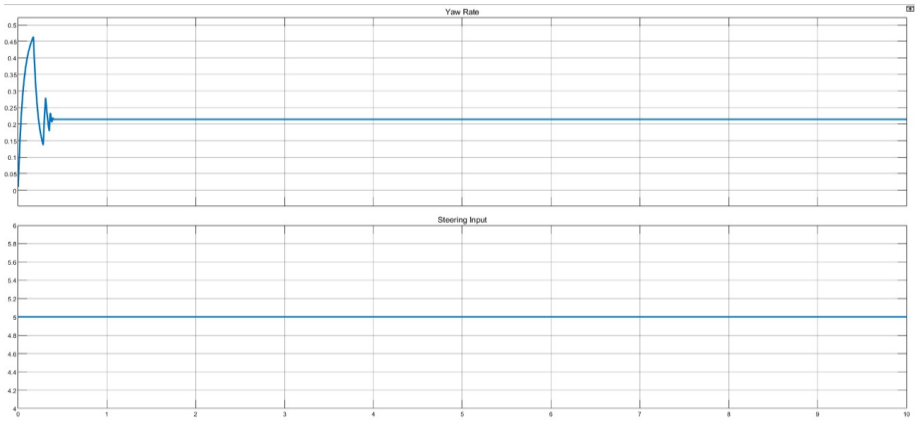
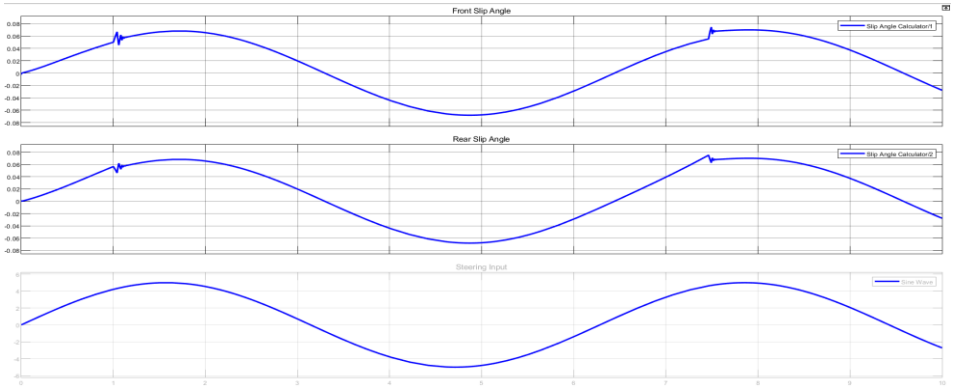


Figure 7 Steering Input of 5 degrees, Outputs: Front & Rear Slip Angles

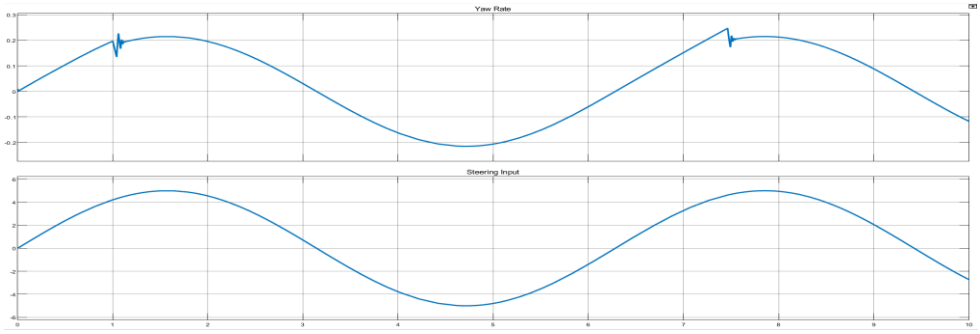


Result 7: Yaw Rate v. Steering Input at 5 degrees

For this result, the steering input and the speed both are constants. The front and rear slip angles are exposed to the PID controller. For the first parts, both angles try to be equal which is completed in less than one second, hence making the car near the neutral steering. This indicates the robustness and fast response of the controller.

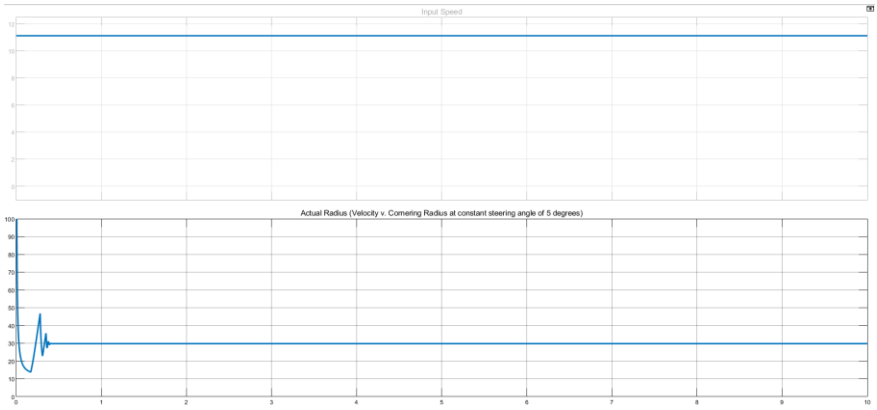


Result 8: Slalom steering input, Outputs: Front and Rear Slip Angles

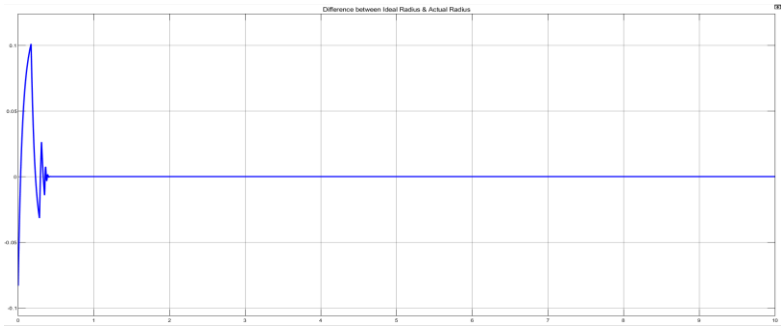


Result 9: Yaw Rate vs Steering input as Slalom

As explained in the previous steering input. The front and rear slip angles are exposed to the PID controller. For the first parts, both angles try to be equal which is completed in less than one second, hence making the car near the neutral steering. A noticeable part in this graph is how the slalom turning radius increases at one part and decreases at the other to make the car in a neutral steering position .This indicates the robustness and fast response of the controller.

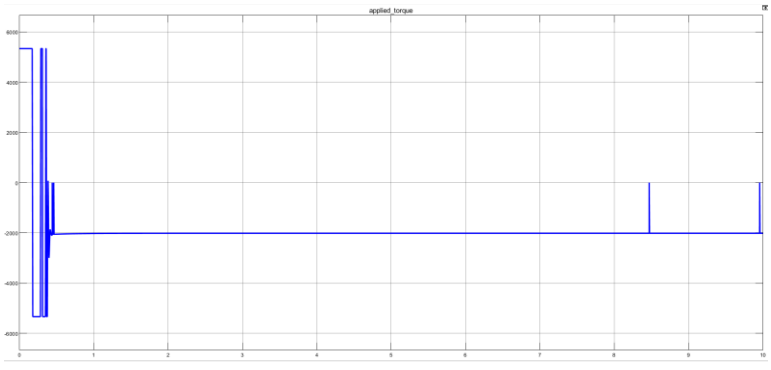


Result 10: Steady-State Response of speed vs Actual Radius with constant steering of 5 degrees



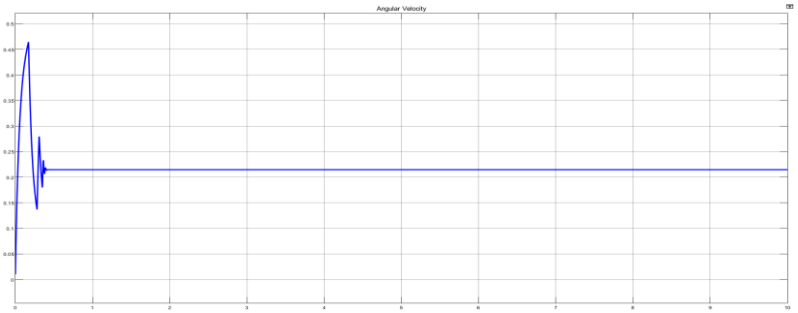
Result 11: Difference between Ideal Radius and Actual Radius after applying the PID controller to the model.

Here the response shows that the PID is working correctly and bringing both to the same value hence controlling the vehicle Yaw Rate by applying the braking force.



Result 12: Applied torque graph.

Here is the graph of the applied torque through the model to make the vehicle yaw control system.



Result 13: Angular Velocity in rad/sec after applying the PID

V. Conclusions and Future Work

Although this model response behave as expected and desired, a 3DOF Model would be more accurate to tackle a yaw control system. The vehicle will be modeled with 4 wheels instead of 2 which will give more range of control and will be more accurate when applying the braking force instead of 2 wheels. However, modeling the vehicle in 3DOF will add more complexity to the model and might add more time to find accurate responses. The PID controller response will be most accurate when compared to the current one as the ideal and actual radius will be impacted by the fact that the model will be simulated in a transient environment and the speed shall not be constant.

A 3DOF model would also allow for the modeling of four wheels instead of just two, which would give a wider range of control and a more accurate representation of the vehicle's behavior when applying braking force as well as we can explore different braking scenarios and even take into consideration the fact of 4 wheel braking system or a two wheel braking system. This would be particularly useful for yaw stability control using differential braking, as it would allow for a more precise control of the vehicle's yaw motion.

However, it is also true that a 3DOF model would be more complex and would require more computational resources and time to find accurate responses. Additionally, the use of a PID controller response would be more accurate when compared to a 2DOF model, as the ideal and actual radius

would be impacted by the fact that the model will be simulated in a transient environment and the speed will not be constant.

there is also the fact we used auto-tuning in order to find the PID parameters, which is not the only method that can be used there are several methods to calculate the PID parameters such as Ziegler-Nichols method, Cohen-Coon method however we focused on the control algorithms and the model itself rather than the PID tunning methodds which could have shown a better results whether in the curve smoothness or decreasing the overshoot.

While a 2-DOF bicycle model is a good starting point for the development of a yaw control system and is more simpler and easier to implement as there are multiple sources and researches that have worked on the same topic, a 3DOF model would be more accurate and appropriate for the task. However, it would also add complexity to the model and require more computational resources which is not easily available.

Moreover, the use of a PID controller on a bicycle model will result in a more stable and controlled response, as shown by the decrease in yaw rate error over time and the smooth transition between the desired yaw rate and the actual yaw rate in the graphical readings. Without a PID controller, the graphical readings will show a larger yaw rate error, larger overshoots, and oscillations resulting in less stable and controlled response.

To summarize, The bicycle model's simplicity allows for fast and efficient calculations, making it well-suited for real-time control systems, in addition when using it with a PID controller which will increase the efficiency of the results , however for more complex system its preferable to use a 3-DOF system to achive the most reliable and accurate response as it will take into consideration the factors that we did not include in the model

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